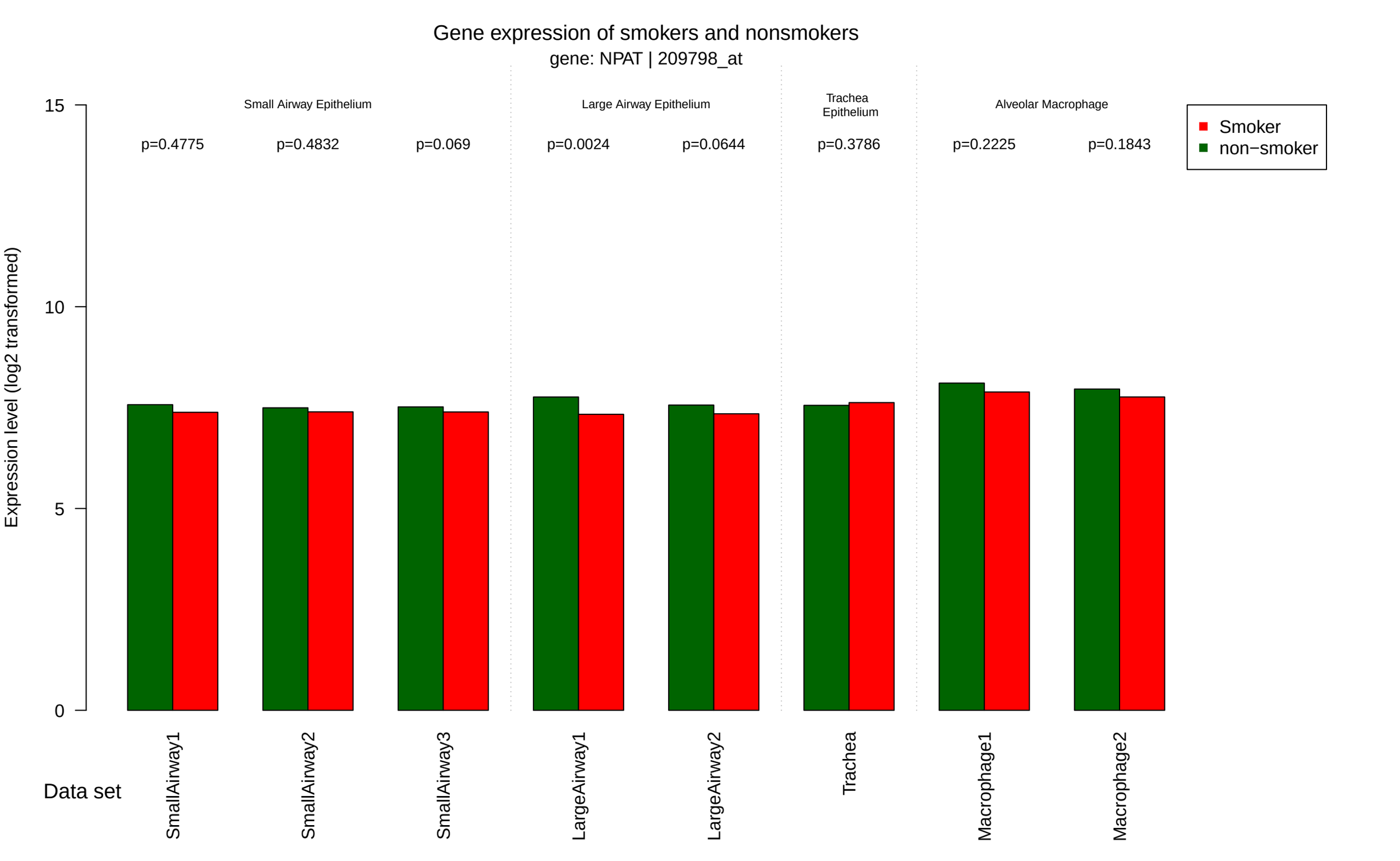
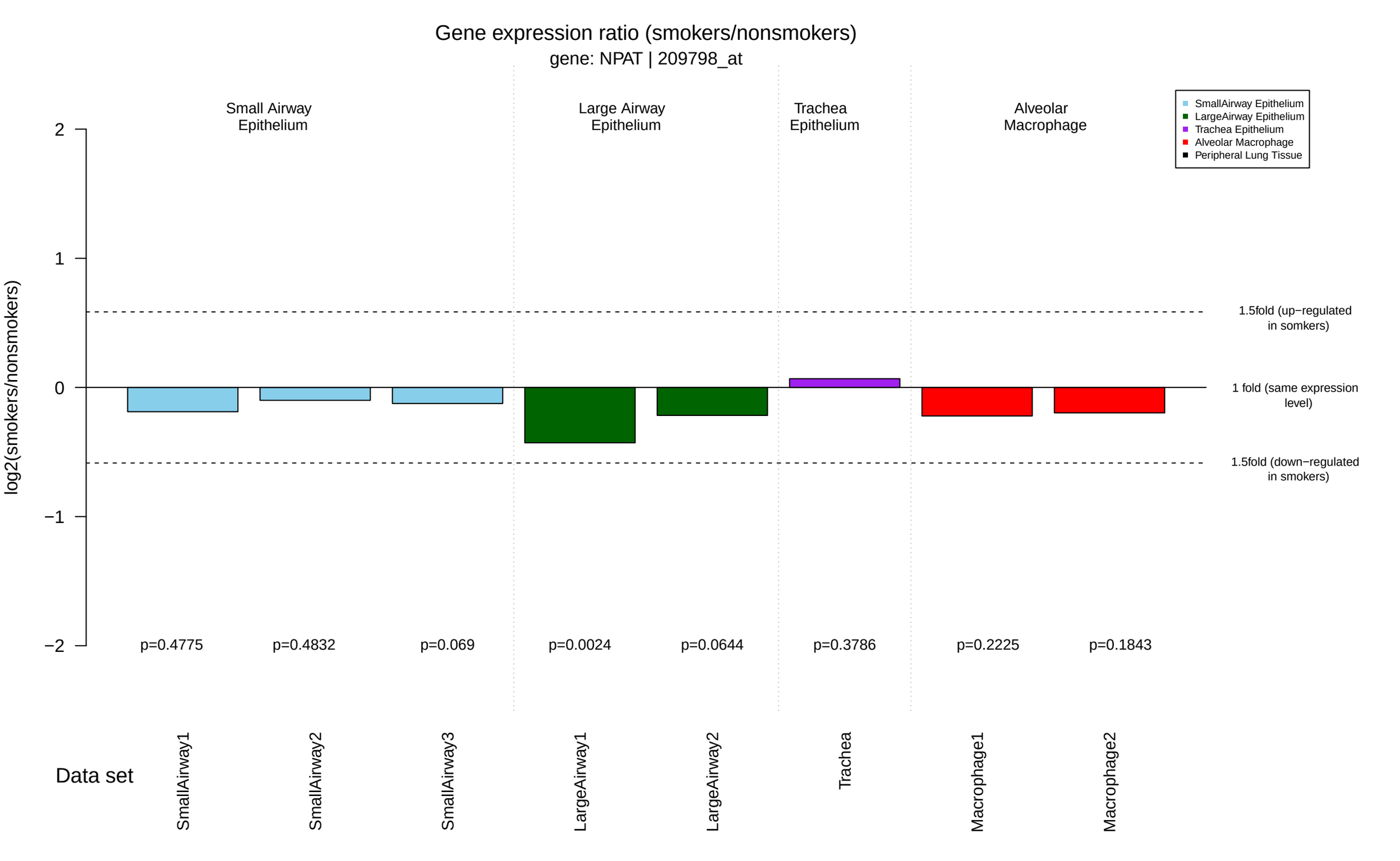
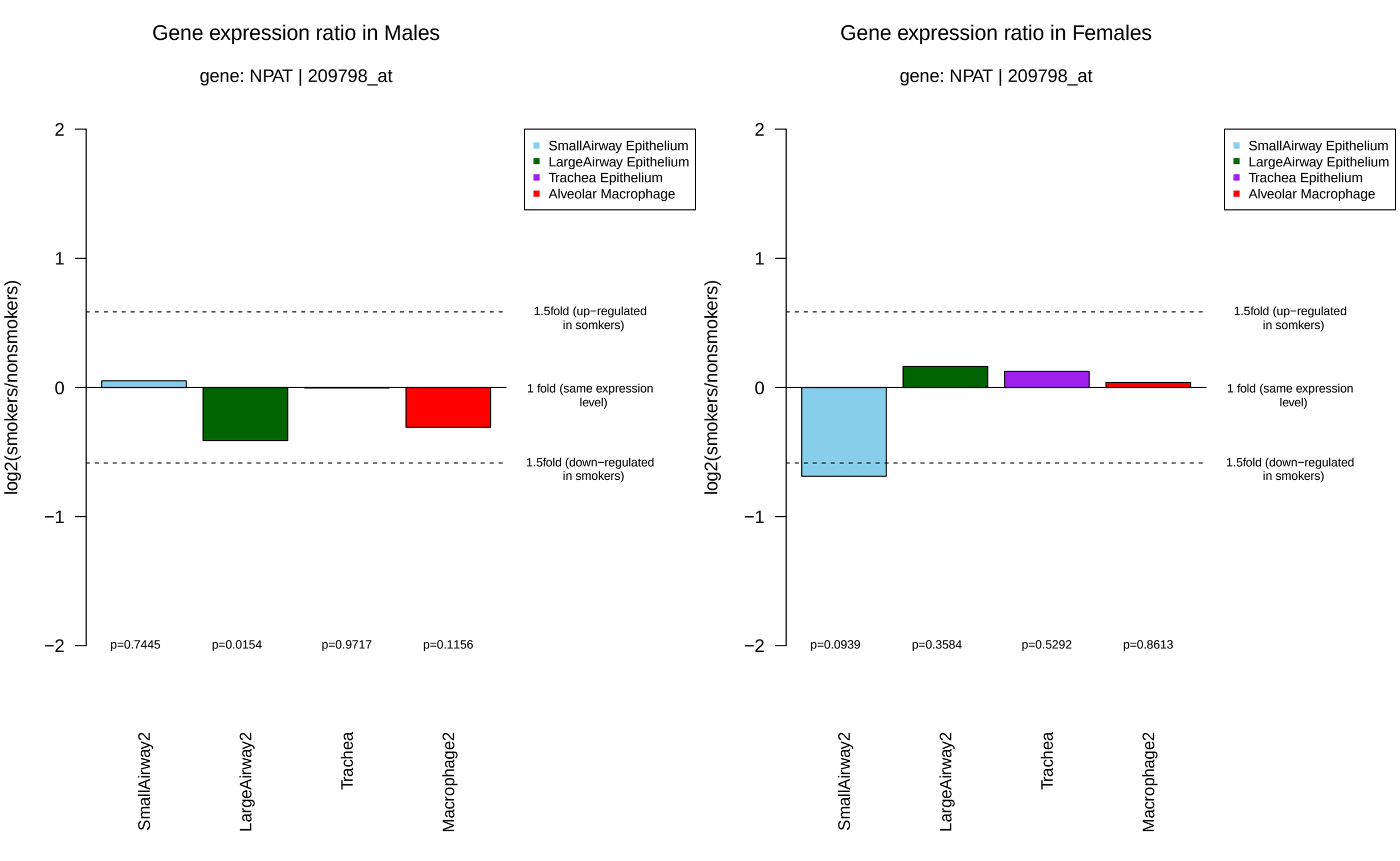
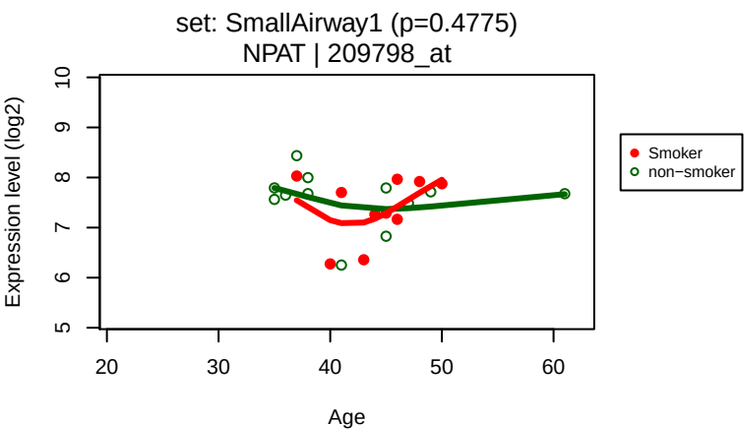
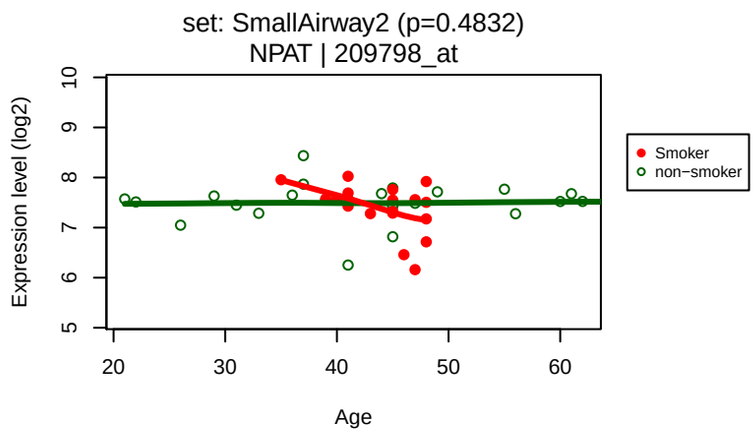
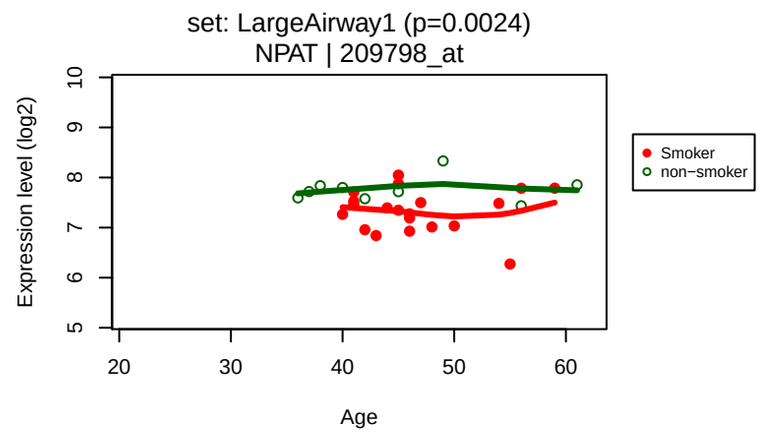

Gene expression of smokers and nonsmokers
gene: NPAT | 209798_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4775
p=0.4832
p=0.069
p=0.0024
p=0.0644
p=0.3786
p=0.2225
p=0.1843
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

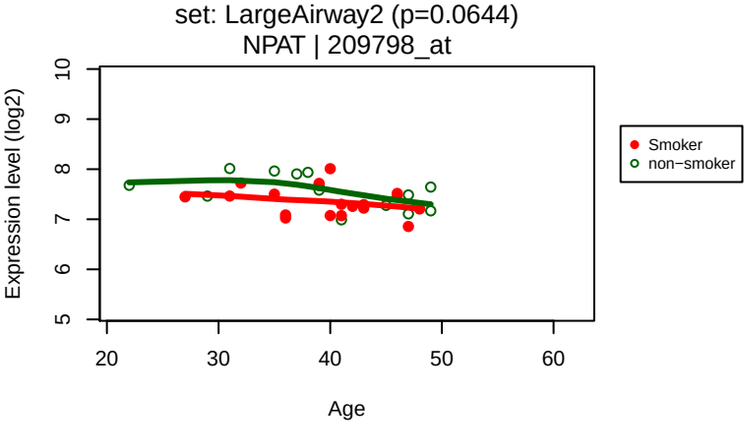

Gene expression ratio (smokers/nonsmokers)
gene: NPAT | 209798_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4775
p=0.4832
p=0.069
p=0.0024
p=0.0644
p=0.3786
p=0.2225
p=0.1843
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

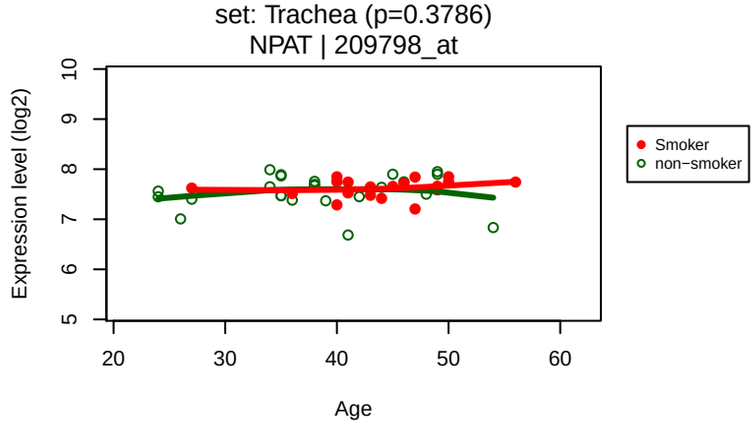

Gene expression ratio in Males
gene: NPAT | 209798_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7445
p=0.0154
p=0.9717
p=0.1156
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPAT | 209798_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0939
p=0.3584
p=0.5292
p=0.8613
SmallAirway2
LargeAirway2
Trachea
Macrophage2

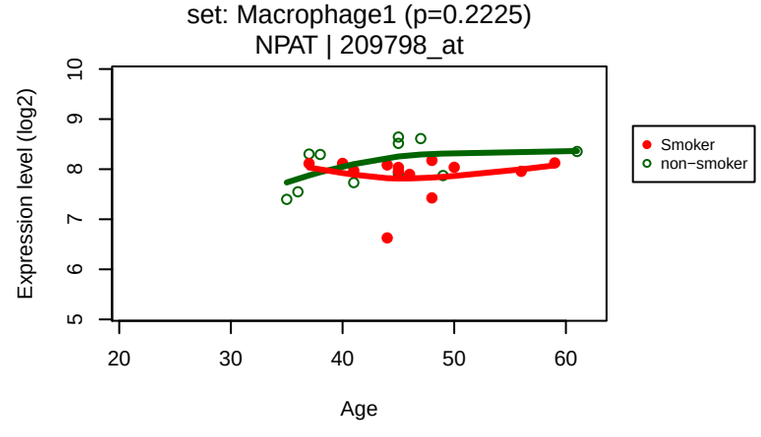

set: SmallAirway1 (p=0.4775)
NPAT | 209798_at
Expression level (log2)
Age
Smoker
non−smoker

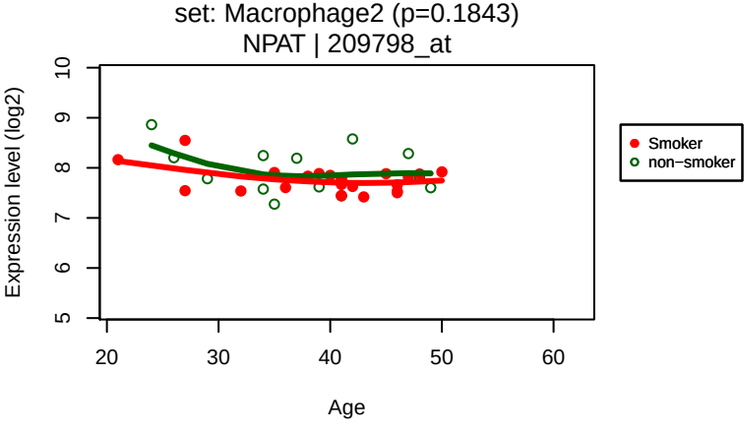

set: SmallAirway2 (p=0.4832)
NPAT | 209798_at
Expression level (log2)
Age
Smoker
non−smoker

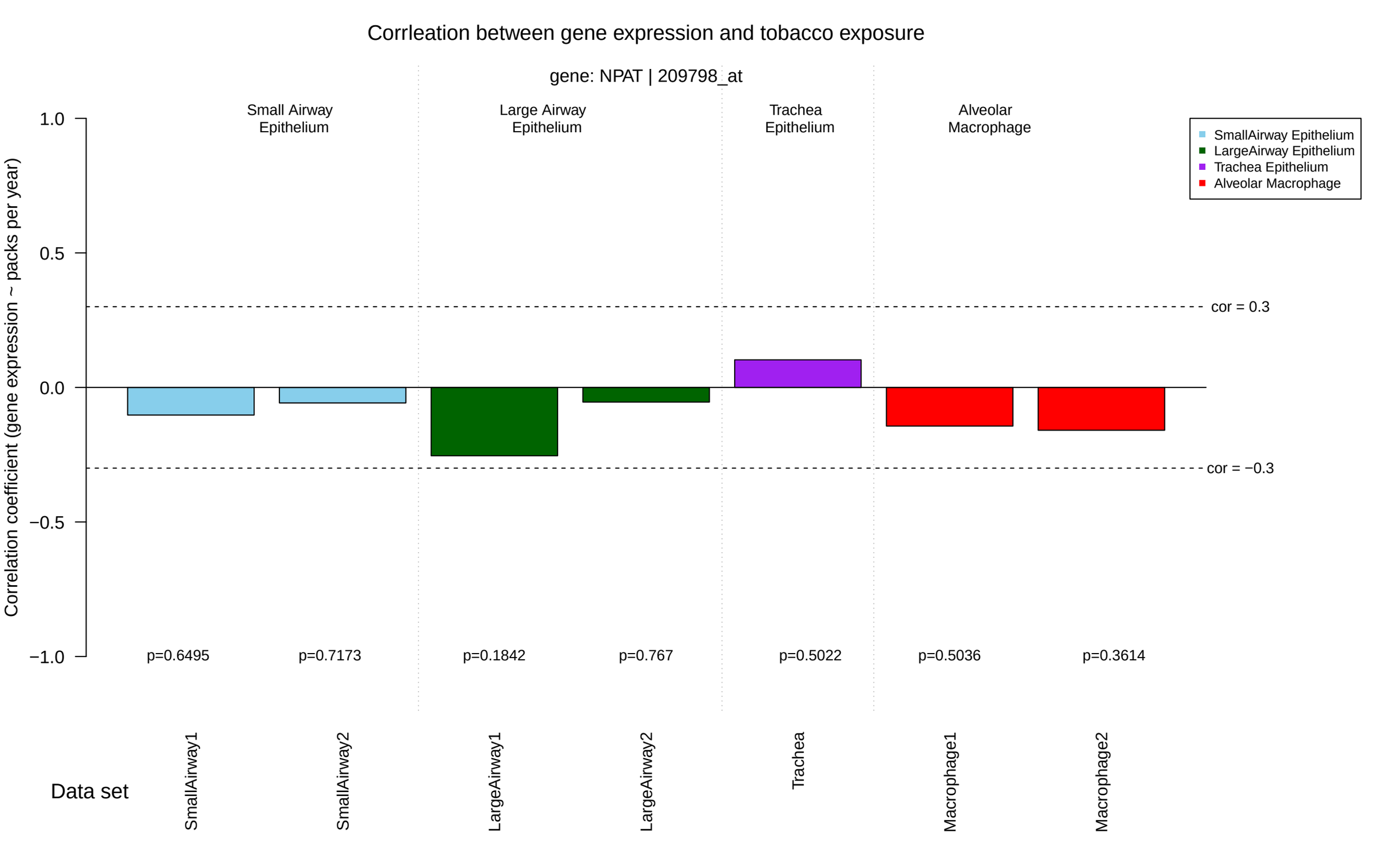

set: LargeAirway1 (p=0.0024)
NPAT | 209798_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0644)
NPAT | 209798_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3786)
NPAT | 209798_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2225)
NPAT | 209798_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1843)
NPAT | 209798_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPAT | 209798_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6495
p=0.7173
p=0.1842
p=0.767
p=0.5022
p=0.5036
p=0.3614
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2